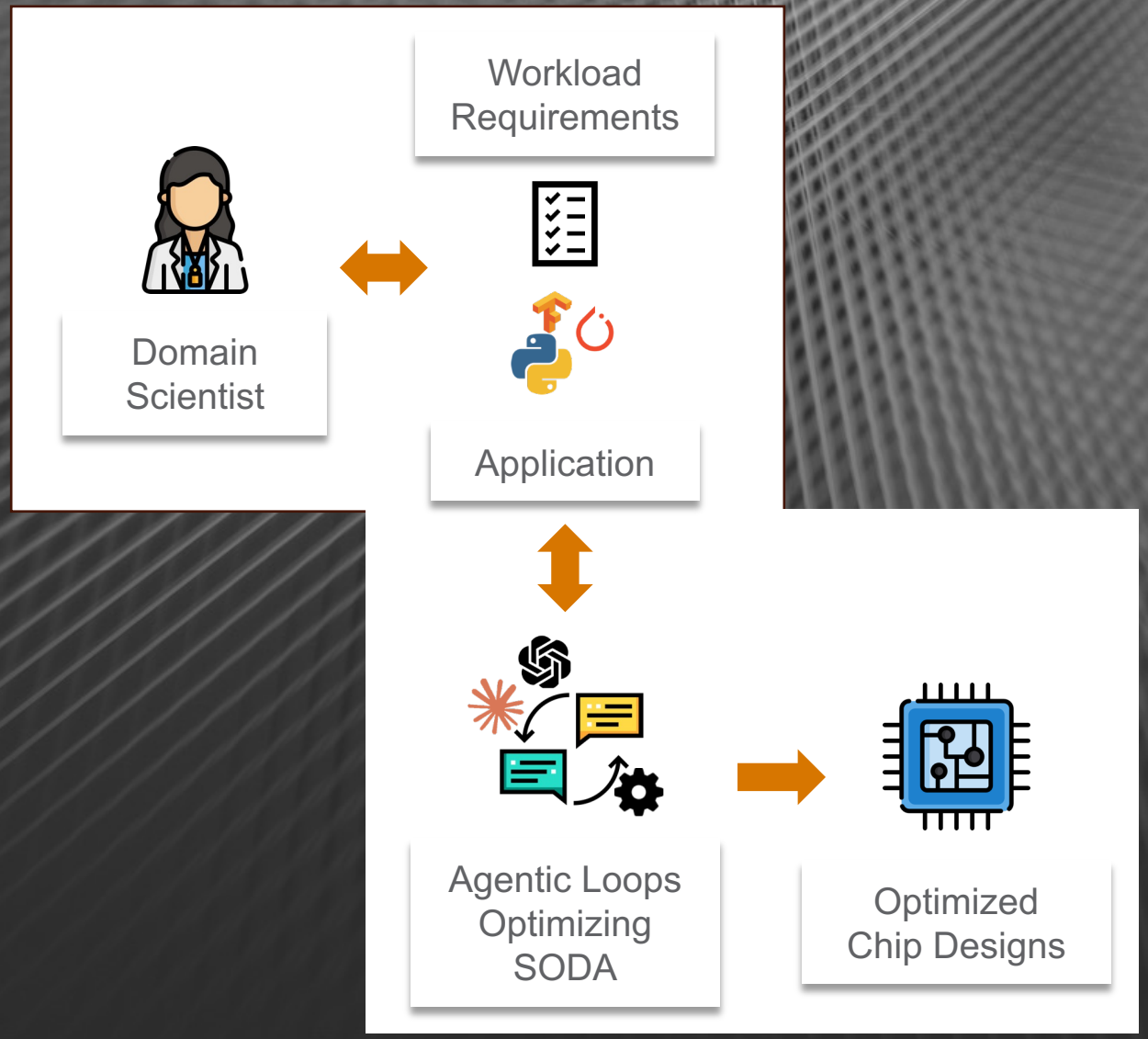
PNNL-SA-220050

CRNCH Summit Panel:

Lab to Fab: Hardware Prototypes (MVPs) as Design Space mid-points

James A. Ang, Ph.D.

Chief Scientist for Computing
Industry Sector Manager: Microelectronics
Pacific Northwest National Laboratory
Richland, WA | USA



Pacific Northwest National Laboratory is operated by Battelle Memorial Institute for the United States Department of Energy under Contract DE-AC05-76RL01830

What is Hardware Prototyping?

- Post Moore's Law Hardware R&D: Bifurcates to Two Tracks
  - Industry Track: products target leading edge technology nodes
    - Slower and Complex Design Cycle
    - Expensive Design Costs: EDA tools, wielded by large teams
    - Costs of each new design are approaching $1B – minimizing technical risk leads to incremental advances
    - → Hardware Prototyping is rare
  - Lab to Fab R&D Track: prototypes target architectural innovation
    - Faster Design Cycle
    - Cheaper Design Costs
    - Opportunities for leap-ahead advances
- Design/Fab/Test Prototype Hardware or a Minimum Viable Product (MVP)

Minimum Viable Product

- Capabilities are focused on core functionality to:
  - Test a novel/innovative idea
  - Validate assumptions of design target
  - Attract early adopters
- Provides valuable feedback for future development
  - Learn from early adopters
  - Refine the design target
- MVPs support agile test and evaluation of hardware designs
  - Rapid design cycle
  - Open-source software stack: compiler frameworks, runtimes
  - Open-source benchmarks and reference workflows

Why is Hardware Prototyping Important?

- Architectural Innovations can support dramatic performance shift: incremental advance → game changing advance
- MVP Role: Proof of Concept System Technology Demonstrator
  - Implement core functionality
  - Provide feedback to hardware designers
  - Attract early adopters – especially software developers
  - Support true hardware-software co-design
  - Measurement of metrics that are difficult to simulate or emulate e.g., energy efficiency, cybersecurity performance.

How can we reduce the cost of Hardware R&D?

- Develop, use and integrate open-source EDA tools for hardware design flows
  - Leverage open compiler frameworks
  - Use open data design files to support co-design collaborations
  - Transition to commercial EDA or proprietary PDKs when required
- Leverage and contribute to the open chiplet eco-system
  - Focus efforts on domain-specialized hardware designs
  - Use commodity chiplets for the rest of the SiP design
- Again, shift from design for product or production to design for R&D and *Proof of Concept* system prototypes

Generative AI Driven Co-design for Microelectronics

- U.S. DOE National Labs have multiple efforts in Microelectronics/Chip Design
  - DOE goals in the CHIPS and Science Act
    - Energy Efficiency in Large Scale HPC and AI
    - Low Latency Edge Computing and Autonomy
    - Advances in Instrumentation and Sensors
- Focus on HPC
  - HPC is evolving to converged applications: Modeling and Simulation + AI/ML + Data Analytics + Custom Accelerators + Digital Twins + etc.
  - Moore's Law Performance Improvements Stalled
  - New Performance Improvements from Customization
- Heterogeneous Computing
  - Programming Models
  - Chiplet Ecosystems
  - Democratization of Co-design

GenAI Opportunities for Advanced Computing

- GenAI offers new opportunities to automate and optimize stages across the electronic design automation (EDA) workflow.
  - Capture domain scientist description of scientific discovery technical requirements and constraints and iterate on optimized hardware designs
  - Generation of verification test vectors for proof-of-concept testbeds
  - Improvements in code translation: e.g., Graph Neural Networks to identify IR correlations and potential code optimizations
- Assess Key Challenges: Benchmarking, Data Scarcity, Regulatory Compliance, and Semantic Understanding
  - Start with open or proprietary LLMs: Llama, OpenAI, Anthropic, etc.
  - Open-Source tools enable data collection for training models and evaluation at different stages of the hardware design flow
  - Semantic capture: chain of thought opportunities

Thank you

This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Advanced Scientific Computing Research, in support of the Democratization of Co-Design (DeCoDe) project in the MEERCAT Microelectronics Science Research Center, the End-to-end Co-design (ENCODE) Competitive portfolio project, and PNNL's Data Model Convergence (DMC) and AT-SCALE LDRD projects, under Contract DE-AC05-76RL01830.